

DIVFusion: Darkness-free infrared and visible image fusion

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ABSTRACT

As a vital image enhancement technology, infrared and visible image fusion aims to generate high-quality fused images with salient targets and abundant texture in extreme environments. However, current image fusion methods are all designed for infrared and visible images with normal illumination. In the night scene, existing methods suffer from weak texture details and poor visual perception due to the severe degradation in visible images, which affects subsequent visual applications. To this end, this paper advances a darkness-free infrared and visible image fusion method (DIVFusion), which reasonably lights up the darkness and facilitates complementary information aggregation. Specifically, to improve the fusion quality of nighttime images, which suffer from low illumination, texture concealment, and color distortion, we first design a scene-illumination disentangled network (SIDNet) to strip the illumination degradation in nighttime visible images while preserving informative features of source images. Then, a texture-contrast enhancement fusion network (TCEFNet) is devised to integrate complementary information and enhance the contrast and texture details of fused features. Moreover, a color consistency loss is designed to mitigate color distortion from enhancement and fusion. Finally, we fully consider the intrinsic relationship between low-light image enhancement and image fusion, achieving effective coupling and reciprocity. In this way, the proposed method is able to generate fused images with real color and significant contrast in an end-to-end manner. Extensive experiments demonstrate that DIVFusion is superior to state-of-the-art algorithms in terms of visual quality and quantitative evaluations. Particularly, low-light enhancement and dual-modal fusion provide more effective information to the fused image and boost high-level vision tasks. Our code is publicly available at <https://github.com/Xinyu-Xiang/DIVFusion>.

1. Introduction

Due to technical limitations and the impact of the shooting environment, a single image captured by the same device often fails to provide a comprehensive description of the entire scene. Therefore, image fusion techniques are emerging, which can extract the most meaningful information from different source images and merge it into fused images. Fused images often contain richer information and promote subsequent applications. In the field of image fusion, infrared and visible image fusion is the most widely used image fusion technique [29]. Specifically, visible images contain abundant texture information and are more adaptive to visual perception. By capturing rich thermal radiation information, infrared images provide highlight significant targets such as vehicles, pedestrians, *etc.*, even in low-light or other extremely harsh environments. For these reasons, infrared and visible image fusion can not only reduce data redundancy but also produce high-quality images with significant contrast and abundant texture details. With these advantages, infrared and visible image fusion technology has good application prospects in military surveillance [6], object detection [22], and vehicle navigation [50], *etc.*

So far, considerable methods have been proposed for infrared and visible image fusion, which can be divided into two major categories: traditional methods and deep learning-based methods. The traditional approaches usu-

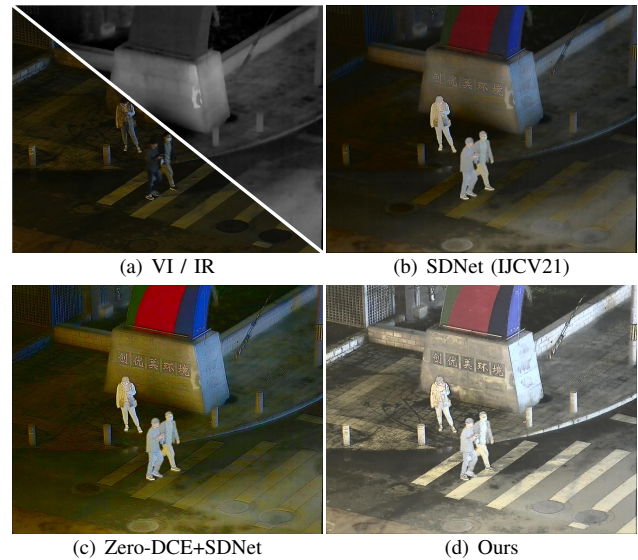


Figure 1: Nighttime infrared and visible image fusion results. Compared with the fusion result of SDNet on original low-light images and the enhanced version by Zero-DCE, our DIVFusion can provide a daytime-like fused image with great visual perception.

ally involve the following three steps. Firstly, a specific transformation is employed to extract features from the source images, which is the feature extraction stage. Then, some fusion strategies are applied to merge these features in the feature fusion phase. Finally, the fused image is reconstructed from the merged features according to the corresponding inverse transformation in the feature recon-

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struction stage. Depending on applied mathematical transformations, traditional infrared and visible image fusion methods can be further divided into five categories including multi-scale transform-based [3, 4], sparse representation-based [46], saliency-based [25], subspace-based [36], and hybrid-based methods [35].

The rise of deep learning has given more possibilities for infrared and visible image fusion, deep learning-based methods can usually achieve more desirable performance exceeding traditional ones [53]. Deep learning-based methods can be further divided into three categories according to the network architecture: CNN-based methods [54], AE-based methods [27], and GAN-based methods [33]. CNN-based methods can achieve feature extraction, feature fusion, and feature reconstruction and obtain unique fusion results through well-designed network architectures and loss functions. Different from CNN-based methods, AE-based methods employ an auto-encoder to complete feature extraction and feature reconstruction, while feature fusion is accomplished by specific fusion strategies. On the basis of CNN-based methods, GAN-based methods introduce generative adversarial mechanisms into the infrared and visible image fusion field. GAN-based fusion methods contain generators and discriminators. Concretely, the discriminator is utilized to constrain the distribution of fusion results generated by the generator to be as close as possible to the source images without supervision.

Although existing deep learning-based methods can effectively integrate vital and complementary information from visible and infrared images, there are still some obstacles that need to be addressed. First, existing methods are all designed for normal light conditions, thus ignoring difficulties of illumination degradation in visible images at night. Specifically, in low-light conditions, previous fusion methods only use infrared information to fill in the scene deficit caused by illumination degradation of the visible image. As a result, the abundant scenario information in nighttime visible images fails to be expressed in fused images, which is deviating from the original intention of the infrared and visible image fusion task. Second, an intuitive solution is to pre-enhance visible images using advanced low-light enhancement algorithms and then merge source images by fusion methods. However, treating image enhancement and image fusion as separate tasks often leads to incompatible problem, which results in poor fusion results as shown in Figure 1(c). Specifically, due to the weak light of the night scene, nighttime visible images have slight color distortion. Low-light enhancement algorithms change the color distribution of the light source and further amplify color distortion in the entire image [45] to some extent. Moreover, during the fusion process, color distortion also occurs in the fused image because the fusion strategy applied in the Y channel changes the saturation distribution in source images as shown in Figure 1(b)&(c).

To address the above-mentioned challenges in infrared and visible image fusion, we accomplish the facilitation of the enhancement task to the fusion task by coupling the vi-

sual enhancement and image fusion techniques in a network. In order to obtain informative fused images with good visual perception, the key of our method is to largely eliminate the incompatibility between two tasks. For this reason, we first devise a scene-illumination disentangled network (SIDNet) by combining the channel attention mechanism and Retinex theory. Specifically, considering the coupling of enhancement and fusion tasks, Retinex theory is applied at the feature level to strip degenerate illumination from mixed features while generating features of enhanced visible images and infrared images. Second, a texture-contrast enhancement fusion network (TCEFNet), containing a gradient retention module (GRM) and a contrast enhancement module (CEM), is designed to achieve feature integration and feature enhancement. GRM maximizes the feature gradient extraction through the first-order and second-order gradient residual streams. CEM considers multi-scale deep features and integrates the attention mechanism to accomplish feature-level contrast enhancement. In addition, it is worth mentioning that we propose a color consistency loss to preserve more visible information in the fused image and also alleviate color distortion. Specifically, we construct an angle loss based on the RGB color space of the visible image, which can effectively reduce the corruption of the color information caused by the weak light in nighttime.

In summary, the main contributions of the proposed method are as follows:

- We propose a novel visually enhanced infrared and visible image fusion framework to enhance visual perception and integrate complementary information, especially in extreme low light conditions.
- A scene-illumination disentangled network (SIDNet) is designed to strip the degenerative illumination features and reinforce unique features of two modalities, to achieve visual enhancement. A texture-contrast enhancement fusion network is constructed to boost contrast as well as texture, and achieves effective complementary information integration.
- We design a color consistency loss to ensure the visual quality of the fused image, which can reduce the color distortion in the fused image and inject more information from the visible domain into the fused image.
- Our fusion results possess brighter scenes and higher contrast without color distortion while obtaining complementary information from source images, as shown in Figure 1(d). Pedestrian detection experiments demonstrate the facilitation of our results in high-level vision tasks.

The rest of this article is organized as follows: Section 2 discusses infrared and visible image fusion methods based on deep learning and Retinex-based image enhancement methods. In Section 3, we describe the problem formulation, network architecture, loss function, and training

details. Section 4 shows the comprehensive qualitative and quantitative evaluation of our DIVFusion, and we compare it with some state-of-the-art methods. Finally, we give the conclusion in Section 5.

2. Related work

In this section, we first briefly review existing fusion methods based on deep learning. Then Retinex theory [19] based image enhancement methods are also introduced.

2.1. Deep learning-based fusion methods

2.1.1. Fusion methods based on CNN

Infrared and visible image fusion algorithms based on convolutional neural network (CNN) utilize elaborate loss functions and network structures to achieve feature extraction, aggregation, and image reconstruction. A representative CNN-based work is PMGI [54], which designs the gradient and intensity ratio in the loss to guide the network to directly generate the final fused image. However, the retention of gradient information is changed by manual adjustment, which causes some loss of texture structure. The above problem was improved in later versions, *i.e.*, SDNet [52]. To inject the contrast information from the infrared image into the fused image to a great extent, a salient target mask is utilized to assist the fusion task in STD-FusionNet [31]. Considering the advantage of multi-scale features in the fusion process, RXDNFuse [27] combines the structural advantages of ResNet [12] and DenseNet [13] to extract different levels of features more comprehensively to achieve an effective fusion. To accomplish the fusion task in different resolutions, Li *et al.* [20] took advantage of the meta-learning and deployed only one CNN model to complete infrared and visible image fusion, which greatly expands the application scope of the fusion model. In order to better combine the fusion task with the high-level vision task, Tang *et al.* [41] applied a novel semantic loss to make fused images friendly for high-level vision tasks. In addition, the Transformer structure is recently introduced similar to CNN, which achieves impressive performance in various vision tasks [8], [2]. Hence, Ma *et al.* [30] designed a general fusion approach based on the Transformer architecture, where the attention-guided cross-domain module can integrate global complementary information. Nonetheless, the illumination condition is also an issue that should not be ignored in the fusion task, PIAFusion [42] takes the illumination into account but the model is too simple to adjust illumination in the complex environment.

2.1.2. Fusion methods based on auto-encoder

With the deepening of research, researchers have proposed tremendous fusion methods based on auto-encoder. Most of them use auto-encoder to extract features from source images and accomplish image reconstruction. The feature fusion process mainly applies manually designed fusion rules. A typical AE-based fusion method, *i.e.*, DenseFuse [21], consists of convolutional layers, fusion layers, and dense blocks. Considering the limited feature extraction

capability of the auto-encoder structure, Li *et al.* further proposed NestFuse [22] and RFN-Nest [23]. The former introduces the nest connection in the network which can extract multi-scale features from the source images. The latter devises a novel detail retention loss function and a feature enhancement loss function to force the network to gain more highly integrated detail features. As multi-scale features cannot eliminate redundant information from source images, Jian *et al.* [15] employed attention mechanisms to concentrate on salient targets and texture details of source images. In order to provide a preliminary exploration of interpretability in the fusion field, DRF [48] decomposes the source images into scene components and attribute components and fuses them separately. Although DRF considers the interpretability of feature extraction, it ignores the interpretability of fusion rules, *i.e.*, hand-crafted fusion rules are not necessarily suitable for merging deep features. Therefore, Xu *et al.* [49] further proposed a learnable fusion rule, which evaluates the significance of each pixel in the feature to the classification result, and then further improves the network interpretability.

2.1.3. Fusion methods based on GAN

Given the powerful ability to estimate probability distributions in an unsupervised manner, the generative adversarial network (GAN) is perfectly suited for unsupervised tasks such as image fusion. FusionGAN [33] is a pioneer in applying GAN to the image fusion field that establishes a generative adversarial framework between the fused image and visible image so that the fused image can obtain texture structure to a greater extent. Nevertheless, a single adversarial game can easily lead to an unbalanced fusion. To improve this problem, Ma *et al.* [32] proposed a dual discriminator conditional generation adversarial network, (*i.e.* DDcGAN [32]) to achieve balanced image fusion, in which both infrared and visible images participate in the adversarial process. Based on DDcGAN, Attention-FGAN [24] additionally employs the multi-scale attention mechanism to preserve the foreground target information of infrared images and rich background detail features of visible images. However, the generative adversarial networks with dual discriminators are not easy to train. On this basis, Ma *et al.* [34] further proposed generative adversarial networks with multi-classification constraints, which can be used to balance the information between the infrared and visible images. Nevertheless, these approaches pursue better visual quality but neglect the facilitation of fusion results on subsequent high-level vision tasks. Liu *et al.* [26] proposed a bilevel optimization formulation for the joint problem of fusion and detection.

However, existing deep learning-based and traditional fusion methods all emphasize the injection and balance of complementary information from the infrared and visible images while ignoring extreme environments, such as the most common illumination degradation. In the night environment, severe dark scenes sharply degrade visible images, resulting in fused images no longer providing good vi-

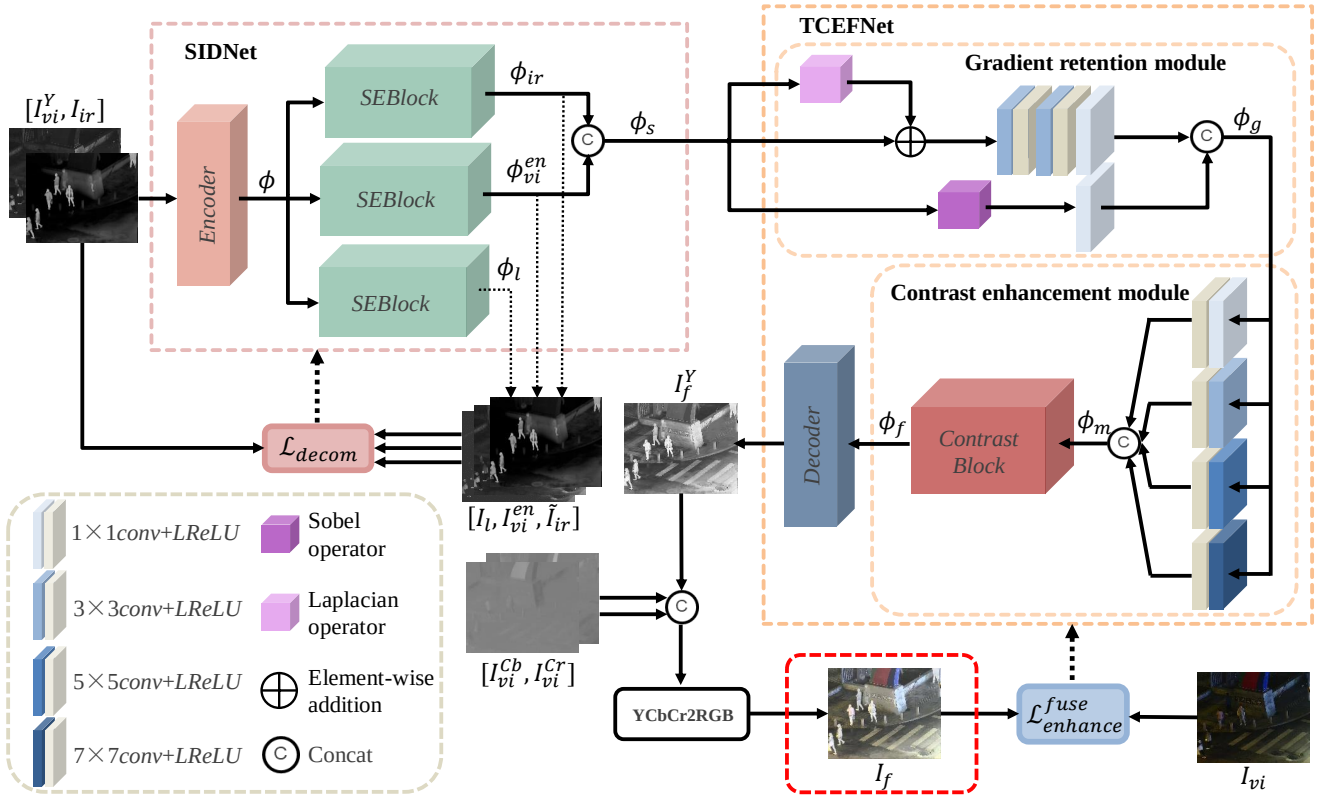


Figure 2: The overall framework of the proposed DIVFusion. SIDNet is a network to separate illumination degradation. TCEFNet integrates and enhances the complementary information of source images.

sual perception. Therefore, corresponding high-level vision tasks may be severely affected. It is worth remarking that nighttime fusion is an important application scenario for infrared and visible image fusion, but it cannot be accomplished by existing fusion methods. To this end, it is urgent to devise a darkness-free infrared and visible image fusion method for extreme conditions such as nocturnal scenarios.

2.2. Retinex-based image enhancement methods

Retinex theory is a calculation theory of color constancy. As a model of human visual perception, it assumes that observed images can be decomposed into reflectance and illumination, denoted as:

$$I = R \cdot L, \quad (1)$$

where R and L represent reflectance and the illumination of the original image, respectively. Reflectance R describes inherent properties of the object that can be considered consistent under any luminance condition. Illumination L depends on the ambient light on the object.

Later on, Retinex models are introduced into the low-light enhancement problem. Jobson *et al.* [17] first used single-scale Retinex (SSR) to strip illumination from the dark image and regard reflectance as the enhanced image. However, the simple decomposition causes severe color distortion. MSRRCR [18] is proposed to improve the color distortion problem of the previous methods. However, final

enhancement results of above methods often look unnatural and over-enhanced in some places. Wang *et al.* [44] proposed to jointly enhance contrast and preserve the naturalness of illumination. In order to deal with less-effective image decomposition and slow convergence, Hao *et al.* [11] proposed the semi-decoupled decomposition model which estimates R and L in a more sensible approach.

With the emergence of deep learning, the low-light enhancement community also tries to leverage the network to estimate reflectance and illumination maps. RetinexNet [45] combines Retinex theory to design a deep network, which contains a decomposition module and an illumination adjusting module. Based on RetinexNet, Zhang *et al.* developed KinD [56], which additionally designs a novel network to achieve reflection recovery. In order to improve the flexibility of KinD in adjusting light levels, they [55] revised KinD and proposed a multi-scale illuminance attention module. Moreover, Zhang *et al.* [51] provided an effective histogram equalization prior to complete efficient decomposition in an unsupervised manner. In particular, they regularize the similarity between histogram equalization-enhanced images and restored images. Based on the above prior, we employ Retinex theory at the feature level to strip the degenerate illumination, which contributes to achieving image enhancement.

3. Method

In this section, we introduce our DIVFusion in detail. Firstly, the problem formulation is presented in Section 3.1. Then, we describe the scene-illumination disentangled network (SIDNet) for removing the illumination degradation and the texture-contrast enhancement fusion network (TCEFNet) for feature enhancement and fusion in Section 3.2. Finally, in Section 3.3, we present specific loss functions for training SIDNet and TCEFNet.

3.1. Problem formulation

When illumination degradation occurs in nighttime visible images, the problem of nighttime infrared and visible image fusion tasks can be divided into two sub-problems: visible image enhancement and infrared and visible image fusion. However, there is a serious incompatible problem in the simple combination of existing low-light enhancement algorithms and fusion algorithms. Hence, how to jointly model the enhancement and fusion tasks and bridge the gap between the two tasks becomes the key to nighttime infrared and visible image fusion. Based on the above problems, we design SIDNet and TCEFNet jointly to minimize the gap between low-light enhancement and image fusion. The former is used to strip away the illuminance component of the degraded visible image at the feature level. Concretely, in SIDNet, the auto-encoder and channel attention mechanism are employed to reconstruct two source images, which preserve the useful information of source images and promote the subsequent fusion task. The latter achieves feature integration and enhances the overall visual perception from the aspect of texture and contrast. The overall framework is schematically illustrated in Figure 2.

More specifically, given a pair of strictly registered infrared image I_{ir} and visible image I_{vi} . Both of them are concatenated in channel dimensions and fed into the encoder $E(\cdot)$ of SIDNet which is illustrated in Figure 3 in order to obtain mixed features ϕ with two domain information. The above process can be formulated as follows:

$$\phi = E(I_{vi}^Y, I_{ir}), \quad (2)$$

where I_{vi}^Y denotes the Y channel of the visible image. Secondly, mixed features with two domain information are fed into three squeeze and excitation blocks (SEblock). SE-blocks aim to select domain-specific feature representations of the infrared and visible images from mixed features. Moreover, another role of SIDNet is to separate the illumination component of visible images. Hence, we employ Retinex theory at the feature level to disentangle the visible image into the degenerative illuminance features and the enhanced visible features. Then, three SEblocks combined with Retinex theory can finally obtain infrared features, degenerative illuminance features, and enhanced visible features. The whole process can be represented as:

$$\{\phi_l, \phi_{vi}^{en}, \phi_{ir}\} = \{SE_l(\phi), SE_{vi}^{en}(\phi), SE_{ir}(\phi)\}, \quad (3)$$

where $SE_l(\cdot)$, $SE_{vi}^{en}(\cdot)$, $SE_{ir}(\cdot)$ represent SEblocks which can obtain degenerative illuminance features ϕ_l , enhanced

visible features ϕ_{vi}^{en} , and infrared features ϕ_{ir} , respectively. In the training stage, separated features are fed into three decoders to reconstruct corresponding images. The reconstruction of different images can be expressed as follows:

$$\{I_l, I_{vi}^{en}, I_{ir}\} = \{D_l(\phi_l), D_{vi}(\phi_{vi}^{en}), D_{ir}(\phi_{ir})\}. \quad (4)$$

Among them, $D_l(\cdot)$, $D_{vi}(\cdot)$, $D_{ir}(\cdot)$ represent the illuminance reconstruction decoder, enhanced visible decoder, and infrared decoder, respectively. I_l , I_{vi}^{en} , I_{ir} represent the degenerative illuminance component of the visible image, the enhanced visible image on the Y channel, and the reconstructed infrared image, respectively. It is worth mentioning that only enhanced visible features ϕ_{vi}^{en} and infrared features ϕ_{ir} are concatenated in channel dimensions as the output of SIDNet, which is defined as:

$$\phi_s = \text{concat}(\phi_{vi}^{en}, \phi_{ir}), \quad (5)$$

where $\text{concat}(\cdot)$ represents the concatenation in channel dimensions.

After that, we also design TCEFNet, which contains two modules, *i.e.*, GRM and CEM. The specific design of GRM is illustrated in Figure 4(a). The output features of SIDNet are directly fed into GRM to reinforce the fine-grained representation of features. We use the Sobel operator and Laplacian operator to jointly enhance mixed features. The process of texture enhancement at the feature level is expressed as:

$$\phi_g = \text{concat}(\text{conv}(\phi_s \oplus \nabla^2 \phi_s), \text{conv}(\nabla \phi_s)), \quad (6)$$

where $\text{conv}(\cdot)$ represents several convolutional operations. ∇ and ∇^2 represent the Sobel operator and Laplacian operator, respectively. ϕ_g is mixed features after texture enhancement. CEM employs different convolutional kernels, *i.e.*, 1×1 , 3×3 , 5×5 , 7×7 , to expand perception fields of the network. After that, features containing multi-scale information are reinforced in the contrast block. The architecture of the contrast block is presented in Figure 4(b). On the basis of the channel attention mechanism, the contrast block calculates the contrast of the feature and assigns higher weights for higher contrast features to achieve contrast enhancement. The specific formulation is as follows:

$$\phi_m = \text{concat}(\text{conv}_{n \times n}(\phi_g)), \quad n \in \{1, 3, 5, 7\}, \quad (7)$$

$$\phi_f = \phi_m \cdot V_c, \quad (8)$$

where $\text{conv}_{n \times n}(\cdot)$ represents the convolutional operation with the kernel size of $n \times n$. The activation vectors V_c are obtained by channel attention mechanism in contrast block. ϕ_f is the output of TCEFNet.

Finally, the fused image I_f^Y is reconstructed by the decoder D . It is important to emphasize that I_f^Y is only the fusion result of the Y-channel enhanced visible image and the infrared image, which owns a bright scene but color information is absent. The reconstruction of fused images can be represented as:

$$I_f^Y = D(\phi_f). \quad (9)$$

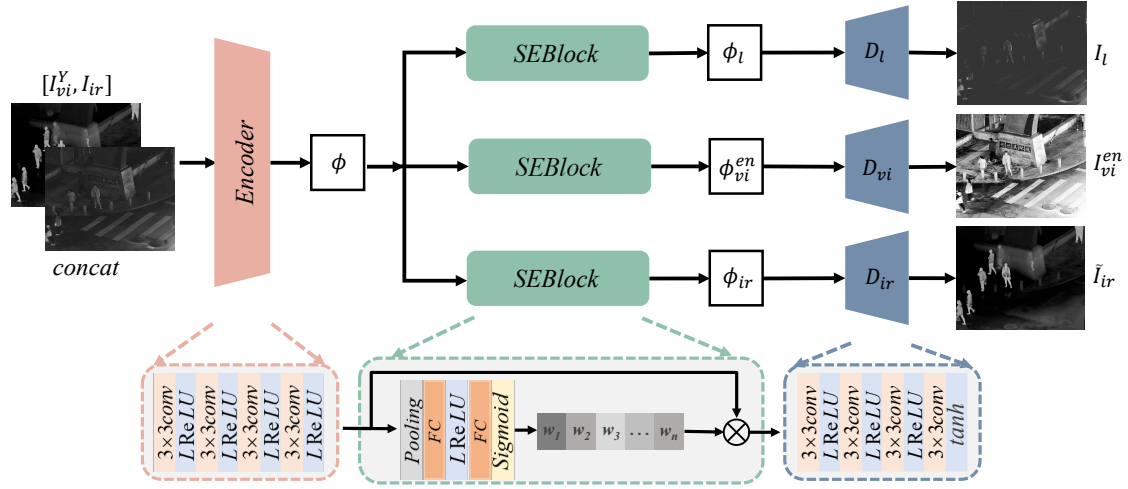


Figure 3: The framework of the scene-illumination disentangled network (SIDNet).

The obtained I_f^Y is first concatenated Cb , Cr channels of the original visible image in channel dimensions to obtain the initial color fused image. Since the Y-channel fusion image I_f^Y only focuses on luminance enhancement and image fusion while ignoring the suitability of chromaticity, the initial fused image has severe color distortion. Then, we convert the color space of the initial fused image to RGB space for correcting the color distribution by using the color consistency loss. Specifically, I_f^Y is adjusted and optimized in RGB space until it matches Cb and Cr channels of the original visible image, which eliminates the color distortion and preserves the scene luminance. The process of converting the initial fused image to RGB space is as follows:

$$I_f = \mathcal{T}(\text{concat}(I_f^Y, I_{vi}^{Cb}, I_{vi}^{Cr})), \quad (10)$$

where $\mathcal{T}(\cdot)$ represents the transfer matrix that converts the image from YCbCr to RGB. I_{vi}^{Cb} and I_{vi}^{Cr} represent Cb and Cr channels of the original visible image, respectively. Then, the mentioned color consistency loss is defined as follows:

$$\mathcal{L}_{color} = \text{angle}(I_f^i, I_{vi}^i), \quad i \in \{R, G, B\}, \quad (11)$$

where $\text{angle}(\cdot)$ represents the discrete cosine operator. More details about the color consistency loss will be demonstrated in Section 3.3.

3.2. Network architecture

As shown in Figure 2, we design the scene-illumination disentangled network and the texture-contrast enhancement fusion network. We introduce the two networks as follows.

3.2.1. Scene-illumination disentangled network (SIDNet)

The specific structure of SIDNet is shown in Figure 3, which consists of an encoder, three parallel SEblocks, and

three parallel decoders. The encoder involves four convolutional layers whose kernel size is 3×3 and the activation function is Leaky Rectified Linear Unit (LReLU). Immediately following are three SEblocks for reinforcing complementary information and removing degenerative illumination. SEblock involves a max-pooling layer, a fully connected layer, a LReLU layer, a fully connected layer, and a sigmoid layer. The decoder consists of four 3×3 convolution layers. The activation function of all layers is LReLU except the last layer, whose activation function is Tanh. It is instructive to note that we only generate enhanced images, degenerative illumination images, and reconstructed infrared images during the training stage. Our model directly generates infrared features and enhanced visible features, and then feeds them into the subsequent network at the inference phase.

3.2.2. Texture-contrast enhancement fusion network (TCEFNet)

TCEFNet consists of two modules, *i.e.*, the gradient retention module (GRM) and the contrast enhancement module (CEM). The specific design of GRM is shown in Figure 4(a). GRM is a variant of resblock, which obtains a main stream and two residual streams. The main stream deploys two common 3×3 convolutional layers with LReLU and a 1×1 convolutional layer. The first residual stream integrates the Sobel operator to retain strong textures of features and the 1×1 convolutional layer to eliminate the difference in channel dimensions. In Figure 4(a), the second residual stream employs the Laplacian operator. Notably, the Laplacian operator is introduced to carefully further extract weak textures of features. Then, the first stage of texture enhancement is accomplished by adding inputs of GRM and outputs of the second residual stream via an element-wise addition. After that, outputs of the main stream and the first residual stream are concatenated in channel dimensions in order to capture deep features with fined-grained details and implement the second stage of texture enhancement.

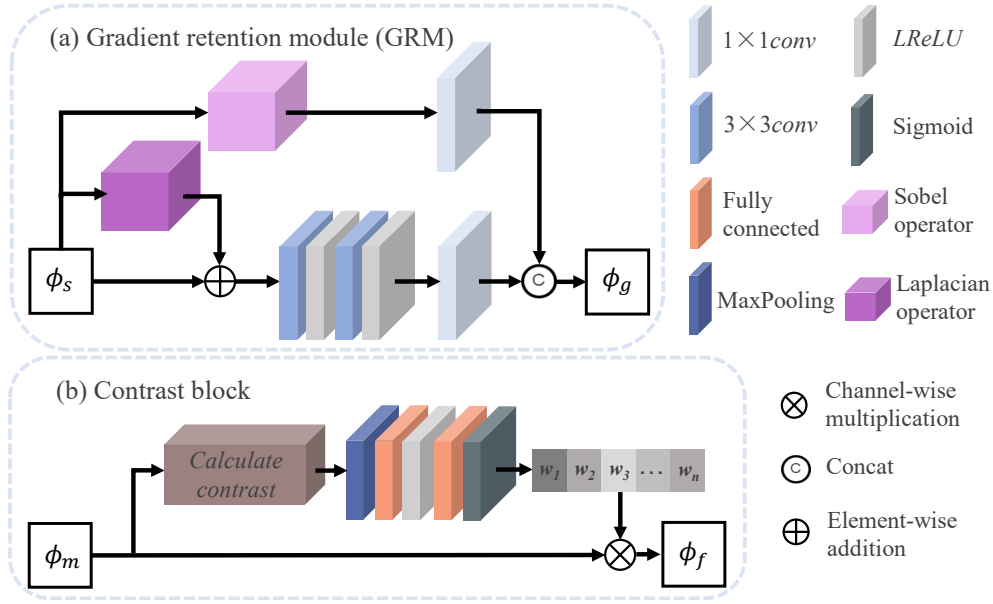


Figure 4: The detailed structure of (a) gradient retention module (GRM) and (b) contrast block.

The enhanced features are fed into CEM which is illustrated in Figure 2. The same-scale convolution hinders the extraction of multi-scale information in different features. Thereby, four convolutional layers with different kernel sizes are utilized to capture multi-scale deep features. In order to reduce information loss in our CEM, different convolutional operations are without pooling layers. The kernel size of convolutional layers is 1×1 , 3×3 , 5×5 , 7×7 , respectively. The activation function of convolutional layers is LReLU. Afterward, obtained features are concatenated in channel dimensions and sent to a contrast block as shown in Figure 4(b). It should be emphasized that we design the contrast block in CEM which enables CEM to boost contrast information at the feature level. In the contrast block, we calculate the standard deviation of each location (i, j) in the residual stream features, which is computed as:

$$\sigma_{ij} = \frac{1}{2r+1} \sqrt{\sum_{-r \leq p, q \leq r} (\phi_m(i+p, j+q) - \mu_{ij})^2}, \quad (12)$$

where r is the radius of the window. μ_{ij} and σ_{ij} are mean value and standard deviation within a window of radius r centered at (i, j) , respectively. Hence, contrast maps are outputs after calculating the standard deviation. Immediately following are a max-pooling layer, a fully-connected layer with LReLU, and a fully-connected layer with sigmoid. The final outputs of the residual stream are activation vectors which represent contrast weights of features. Subsequently, we multiply features in the main stream and activation vectors of the residual stream via channel-wise multiplication to generate enhanced features.

3.3. Loss function

3.3.1. Decomposition loss

In order to ensure our SIDNet strips the degenerative illumination of visible images at the feature level, we design a decomposition loss function containing the following five terms:

$$\mathcal{L}_{decom} = \lambda_1 \mathcal{L}_{recon}^{ir} + \lambda_2 \mathcal{L}_{recon}^{vi} + \lambda_3 \mathcal{L}_{smooth}^{illu} + \lambda_4 \mathcal{L}_{mc} + \lambda_5 \mathcal{L}_{per}, \quad (13)$$

where $\mathcal{L}_{recon}^{ir}$ and $\mathcal{L}_{recon}^{vi}$ are the infrared and visible image reconstruction loss function, respectively. They force mixed features to contain more complementary information with high fidelity. $\mathcal{L}_{mc}$ and $\mathcal{L}_{smooth}^{illu}$ are the mutual consistency loss and illumination smoothness loss, respectively, which guide SIDNet to generate degenerate illumination components from mixed features. The perceptual loss $\mathcal{L}_{per}$ constrains the generation of enhanced visible features during the training process. $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$ represent the hyperparameters that control the trade-off of each loss term.

Infrared and visible reconstruction loss can drive SIDNet to restore original images as much as possible and generate high-fidelity features with less redundancy, which are defined as:

$$\mathcal{L}_{recon}^{vi} = \|I_{vi}^Y - \tilde{I}_{vi}^Y\|_1, \quad (14)$$

$$\mathcal{L}_{recon}^{ir} = \|I_{ir} - \tilde{I}_{ir}\|_1, \quad (15)$$

where I_{ir} and I_{vi}^Y represent the infrared image and Y channel of the visible image. $\tilde{I}_{ir}$ and $\tilde{I}_{vi}^Y$ denote the reconstructed infrared image and Y channel of the reconstructed visible image. On the basis of Retinex theory, $\tilde{I}_{vi}^Y$ is obtained by pixel-wise multiplying the enhanced visible image on Y

channel I_{vi}^{en} and the degenerative illuminance I_l , as presented in the following equation:

$$\tilde{I}_{vi}^Y = I_{vi}^{en} \cdot I_l. \quad (16)$$

The above two loss functions can only constrain final reconstructed infrared and visible images, but fail to constrain illumination components. Thus, in order to strip degenerate illumination components from original visible images, we further apply the following two loss functions to specifically constrain the generation of illumination components, based on the definition of illumination in KinD [56] and RetinexNet [45], which is expressed as follows:

$$\mathcal{L}_{smooth}^{illu} = \left\| \frac{\nabla I_l}{\max(|\nabla I_{vi}^Y|, \epsilon)} \right\|_1, \quad (17)$$

$$\mathcal{L}_{mc} = \|M \cdot \exp(-c \cdot M)\|_1, \quad M = |\nabla I_l|, \quad (18)$$

where ∇ is the Sobel operator including the x and y directions. ϵ is a small positive constant (0.01 in this paper) in order to prevent the divisor from being zero. These functions reduce the risk of over-smoothing on the structure boundary. c is a parameter (10 in this paper) to control the shape of mutual consistency loss, which aims to increase the moderate gradient part of the image. The organic combination of $\mathcal{L}_{mc}$ and $\mathcal{L}_{smooth}^{illu}$ makes the generated illumination components more physically consistent with the definition.

Due to the lack of ground-truth to guide the training process, it is tough to recover enhanced visible images from low-light visible images through indirect loss functions. Fortunately, features of histogram equalization enhanced images can provide abundant texture and luminance information [51]. Thus, we adopt VGG features to constrain the perceptual similarity between the Y channel of low-light visible images and enhanced images by histogram equalization. The customized perceptual loss is shown as follows:

$$\mathcal{L}_{per} = \|VGG(I_{vi}^{en}) - VGG(I_{vi}^{hist})\|_2^2, \quad (19)$$

where $VGG(\cdot)$ represents the pre-trained VGG-19 model [40]. I_{vi}^{hist} and I_{vi}^{en} represent the visible image on the Y channel after histogram equalization and enhanced visible image on the Y channel, respectively. In the pre-trained VGG-19 network, shallow features are concerned about the texture information, while deep features pay more attention to the semantic information. In order to improve the performance of subsequent high-level vision tasks effectively, we expect enhanced features to contain more deep semantic information. Therefore, we only choose the last two layers of the VGG-19 network to implement deep feature extraction, *i.e.*, extracting features from layers *conv4_1* and *conv5_1* to jointly measure the feature similarity.

3.3.2. Enhancement-fusion loss

We expect to train an optimized TCEFNet for image fusion, feature enhancement, and color equalization. The

exact definition of the enhancement-fusion loss is expressed as follows:

$$\mathcal{L}_{enhance}^{fuse} = \alpha_1 \mathcal{L}_{tex} + \alpha_2 \mathcal{L}_{int} + \alpha_3 \mathcal{L}_{color}, \quad (20)$$

where $\mathcal{L}_{tex}$ represents the texture loss, aiming to absorb more texture detail information from source images to the fused image. The intensity loss $\mathcal{L}_{int}$ constrains the fused image to preserve the significant target information of the infrared image. $\mathcal{L}_{color}$ is the color consistency loss, which aims to mitigate the color distortion during the enhancement and fusion processes. $\alpha_1, \alpha_2, \alpha_3$ are hyper-parameters that are used to strike a balance between the above loss.

The texture loss forces more texture details to transfer from source images to fusion results, which is defined as:

$$\mathcal{L}_{tex} = \frac{1}{HW} \left\| |\nabla I_f^Y| - \max(|\nabla I_{vi}^Y|, |\nabla I_{ir}|) \right\|_1, \quad (21)$$

where H and W are the height and width of an image, respectively. $|\cdot|$ denotes the absolute value operation. The texture loss guides the fused image to retain more high-frequency details.

The goal of the intensity loss function is to make the fused image obtain more salient target information of the infrared image, which is defined as:

$$\mathcal{L}_{int} = \frac{1}{HW} \|I_f^Y - I_{ir}\|_1. \quad (22)$$

Since nighttime images are easily affected by different color light, image enhancement methods are prone to suffer from color distortion and over-amplify the color information of the original light source in the scene. Although the scene brightness can be improved, the color distortion is extremely serious. Considering that the Euclidean distance in RGB space does not reflect color distortion effectively [57], we choose the discrete cosine distance to constrain the color distribution of the fused image. This constraint forces the color distribution of the fused image to be closer to the color of the original visible image distribution. Furthermore, we hope the intensity of the visible image can be injected into the fused image through discrete cosine similarity when removing the color distortion. Overall, the designed color consistency loss is formulated as follows:

$$\mathcal{L}_{color} = \frac{1}{HWC} \sum_{i \in \zeta} \sum_{k=1}^K \angle(I_{vi}^i, I_f^i), \quad \zeta = \{R, G, B\}, \quad (23)$$

where C indicates channel numbers of the image. $\angle(\cdot, \cdot)$ illustrates the pixel-wise calculation of the discrete cosine similarity between the fused image and the original visible image in R, G, B channels. K denotes the number of pixels in the image. i is the element of R, G, B. The color consistency loss is designed for two purposes. On the one hand, the color distortion caused by enhancement can be well reduced in the fused image. On the other hand, the visible information can be aggregated into the fused image from R, G, B channels, so that the fused image can also capture more scene detail information.

4. Experimental results

In this section, we first describe the experimental configuration and details of network training. Then we perform fusion comparison, two-stage fusion comparison, and also generalization experiments to illustrate the superiority of our approach. In addition, some ablation studies are conducted to verify the effectiveness of our specific designs. Finally, pedestrian detection is performed to illustrate the facilitation of our algorithm for high-level vision tasks.

4.1. Experimental configurations

We perform qualitative and quantitative experiments on LLVIP dataset [14] to comprehensively evaluate our DIV-Fusion. LLVIP dataset contains aligned infrared and visible images for low-light vision, which are captured on the road at night. In addition, we also select MSRS [42] and TNO [43] datasets to illustrate the generalization ability of our method. The infrared and visible image pairs in MSRS dataset both contain daytime and nighttime scenarios with a spatial resolution of 480×640 . TNO dataset includes multispectral nighttime imagery of various military-relevant scenarios and all images in this dataset are grayscale versions. 50 typical infrared and visible image pairs from LLVIP dataset are used for illustrating the superior performance of the proposed method. In addition, we select 50 typical nighttime image pairs from MSRS dataset and 30 representative image pairs from TNO dataset for generalization analysis. Our results are compared with seven state-of-the-art (SOTA) fusion algorithms, including a traditional method, *i.e.*, GTF [28], two AE-based methods, *i.e.*, DenseFuse [21] and NestFuse [22], two GAN-based methods, *i.e.*, FusionGAN [33] and GAN-McC [34], and two CNN-based methods, *i.e.*, SDNet [52] and U2Fusion [47]. We also design a two-stage fusion experiment to demonstrate the advantages of our method in eliminating incompatibility between enhancement and fusion. In other words, we compare our method with combinations of SOTA low-light enhancement methods and fusion methods to evaluate the fusion performance and observe how significant the enhancement process is for fusion results. Three representative methods, *i.e.*, ZeroDCE [9], KinD [56], and EnlightenGAN [16], are applied to pre-enhance the low-light visible images. All the above image fusion algorithms as well as low-light enhancement algorithms are publicly available, and we set parameters of the original paper.

For quantitative evaluation, six metrics are selected to objectively assess the fusion performance, including average gradient (AG) [5], correlation coefficient (CC) [7], entropy (EN) [39], the sum of correlations of differences (SCD) [1], standard deviation (SD) [37], visual information fidelity (VIF) [10]. AG reflects the richness of the image texture information. CC measures the degree of linear correlation between the fused image and source images. EN evaluates the amount of information contained in the fused image from the perspective of the information theory. SCD reflects the level of correlation between information transmitted to the fused image and corresponding source images. SD

statistically indicates the distribution and contrast of the fused image. VIF measures the fidelity of information from the perspective of human visual perception. Moreover, a fusion algorithm with larger AG, CC, EN, SCD, SD, and VIF indicates better fusion performance.

4.2. Implementation details

We select 240 pairs of nighttime infrared and visible images from LLVIP dataset to train our SIDNet and TCEFFNet. It is worth mentioning that we crop these images into a large number of image patch pairs for more training data. All images are normalized to $[0, 1]$ before being fed into networks.

We train our SIDNet and TCEFFNet separately since our method is a two-stage model. In the first stage, we train SIDNet with batch size set to 5 and patch size set to 224×224 . After that, output features from the previous network are applied as the input of TCEFFNet. The batch size in TCEFFNet is set to 10, and the patch size is set to 64×64 . Besides, parameters in all networks are updated by the adam optimizer with the weight decay equal to 0.0001. The initial learning rate is set to 10^{-4} , which decreases to 10^{-5} after 50 epochs and then to 10^{-6} after 100 epochs. In addition, $\lambda_{(\cdot)}$ of Eq. (13) are set as follows: $\lambda_1 = 2000$, $\lambda_2 = 1000$, $\lambda_3 = 7$, $\lambda_4 = 9$, and $\lambda_5 = 40$. $\alpha_{(\cdot)}$ of Eq. (20) are set as follows: $\alpha_1 = 200$, $\alpha_2 = 2.5$, and $\alpha_3 = 1.1$. The whole network is trained with the Tensorflow framework on NVIDIA TITAN RTX GPU and 3.5 GHz Intel Core i9-9920X CPU.

4.3. Fusion performance analysis

To illustrate the advantages of the proposed method, we conduct comprehensive comparisons of fusion performance with seven SOTA fusion methods on LLVIP dataset.

4.3.1. Qualitative results

In order to visually observe the fusion performance of different algorithms on LLVIP dataset, we select two pairs of infrared and visible images (named #010064 and #060193, respectively). The visualized results are presented in Figure 5.

Due to the problem of illumination degradation in nighttime images, a superior nighttime fusion algorithm should aggregate meaningful information from source images and provide a bright scene with high contrast. As shown in the red box of the #010064 image, fusion results of FusionGAN, NestFuse, and SDNet are relatively blurred, missing a large amount of scene information. Although GTF, GANMcC, DenseFuse, and U2Fusion have more texture information, the overall scene submerges into the darkness. These methods fail to provide a pleasing visual perception. On the contrary, our fusion results obtain bright scenes and abundant texture even in the darkness. It is worth emphasizing that seven SOTA methods cannot show rich details in the dark regions, while our method is prone to capture the texture information regardless of darkness. We zoom in dark areas with the green box in the #010064 image and the red box in the #060193 image to illustrate this phenomenon. As shown in the #060193 image, although GANMcC and NestFuse can

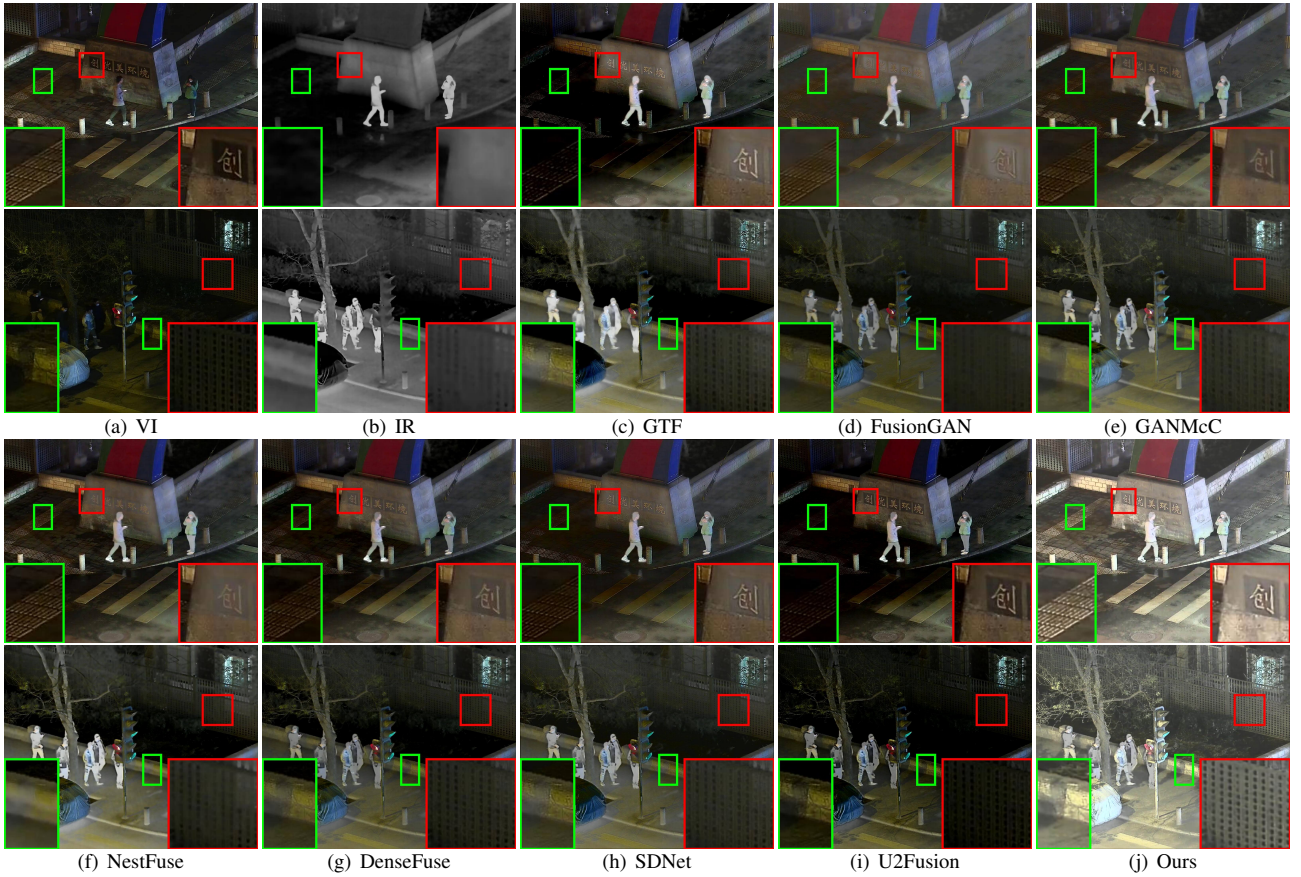


Figure 5: Vision quality comparison of our method with seven SOTA fusion methods on #010064 and #060193 images from LLVIP dataset. For clearer comparison, regions of abundant textures are zoomed in the red box and the green box.

also have a better overall visual perception, texture details cannot be presented clearly, such as the fence region. The fused image generated by GTF is also relatively bright but overly inclines to the infrared image. By contrast, our result contains salient targets and abundant texture information while providing a daytime-like scene.

4.3.2. Quantitative results

We further conduct quantitative comparisons on 50 image pairs from LLVIP dataset to verify the effectiveness of our method, which is presented in Figure 6. It can be seen that our method ranks first in all five metrics and third in the CC metric. The best AG metric indicates that our results have more texture details. The optimal result on the SCD metric demonstrates that our results contain more realistic information. The best SD metric shows that our fused images have richer contrast information which coincides with the visualization results. The presence of enhancement and fusion coupling effects allows the VIF metric of our method to outperform other SOTA methods. Our method trails DenseFuse and GANMcC by a narrow margin in the CC metric, but our fusion result is brighter and the scene is clearer than that of other methods. Our method boosts the overall brightness of scene, which leads to a reduction in the linear correlation between the fused image and source

images. Hence, it is excusable that the CC metric ranks third.

In conclusion, our DIVFusion is fully capable of excavating valid information in dark scenes and integrating it into fused images with the help of feature-level illumination stripping in SIDNet. Thereby, our method has more advantages than other SOTA methods and obtains high-quality fused images.

4.4. Two-stage fusion performance analysis

To illustrate that our DIVFusion can exploit the process-output of low-light enhancement to promote fusion performance, we implement two-stage fusion performance comparisons. Specifically, three representative low-light enhancement algorithms, *i.e.*, Zero-DCE [9], KinD [56], and EnlightenGAN [16] are selected to prioritize Y channel of nighttime visible images for enhancement. Next, infrared and enhanced visible images are fused by seven SOTA fusion methods mentioned above. In addition, the fusion process is performed only on the Y channel, and the chromatic information all comes from visible images.

4.4.1. Enhance by Zero-DCE

As SOTA fusion methods cannot deal with illumination degeneration, a low-light enhancement method (Zero-DCE) is first utilized as a pre-enhance operation for nighttime

Infrared and visible image fusion

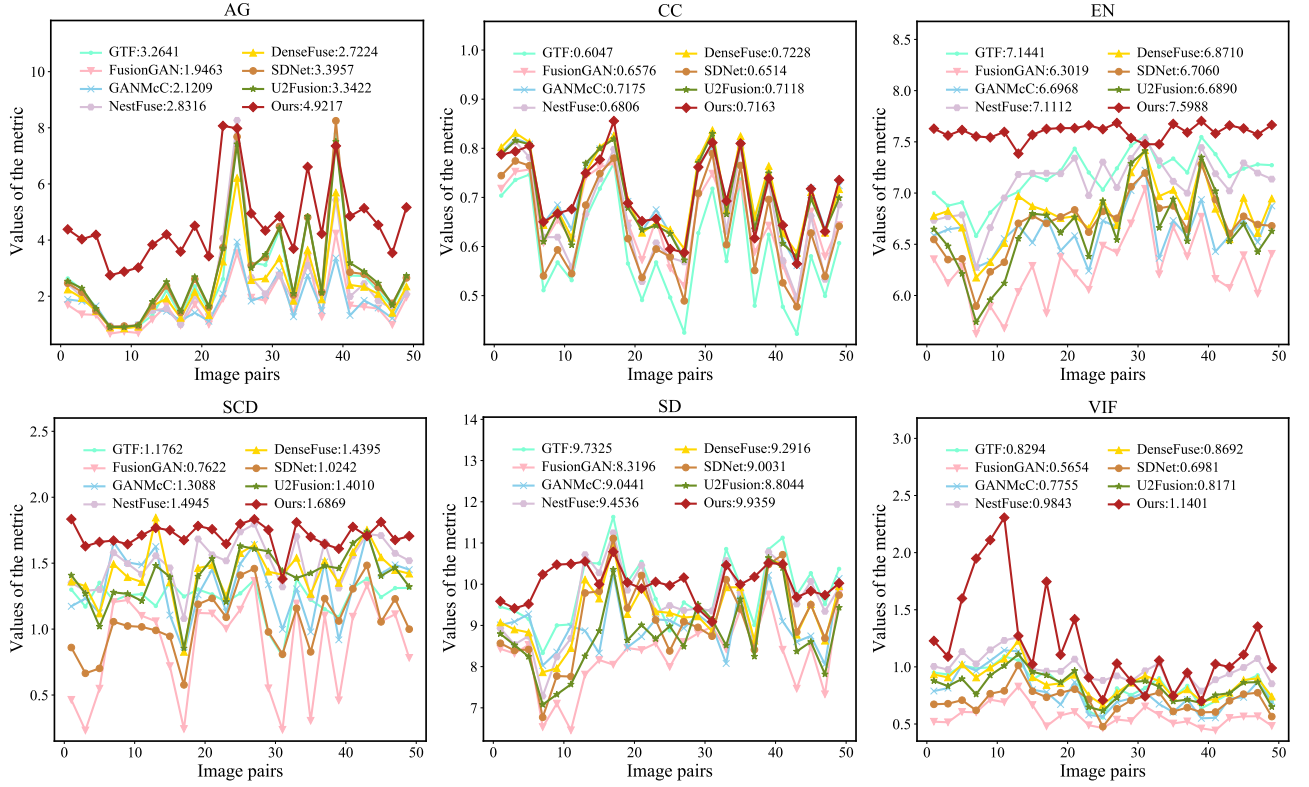


Figure 6: Quantitative results of six metrics, *i.e.*, AG, CC, EN, SCD, SD, and VIF, on 50 image pairs from LLVIP dataset. Seven SOTA methods are used for comparison.

Table 1

Quantitative results of two-stage fusion results on LLVIP dataset. We select six metrics including AG, CC, EN, SCD, SD, and VIF to evaluate ours with seven SOTA methods on 50 image pairs selected from LLVIP dataset. mean and standard variation, **Bold**: best. Underline: second best.

Enhancement Method: Zero-DCE								
	GTF	FusionGAN	GANMcC	NestFuse	DenseFuse	SDNet	U2Fusion	Ours
AG	4.607 ± 2.341	2.796 ± 1.255	2.739 ± 1.071	4.539 ± 2.366	3.541 ± 1.680	4.635 ± 2.427	<u>4.803 ± 2.199</u>	4.922 ± 1.479
CC	0.603 ± 0.107	0.635 ± 0.078	0.697 ± 0.085	0.663 ± 0.078	<u>0.705 ± 0.088</u>	0.650 ± 0.100	<u>0.687 ± 0.093</u>	0.716 ± 0.080
EN	<u>7.166 ± 0.281</u>	6.461 ± 0.274	6.774 ± 0.204	6.804 ± 0.397	6.938 ± 0.248	6.917 ± 0.281	7.150 ± 0.218	7.599 ± 0.075
SCD	1.143 ± 0.205	0.712 ± 0.463	1.146 ± 0.362	0.992 ± 0.414	1.271 ± 0.253	1.103 ± 0.269	<u>1.450 ± 0.214</u>	1.687 ± 0.143
SD	9.770 ± 0.763	8.581 ± 0.724	9.542 ± 0.519	8.994 ± 0.882	9.493 ± 0.667	9.300 ± 0.869	<u>9.980 ± 0.705</u>	<u>9.936 ± 0.437</u>
VIF	0.811 ± 0.167	0.570 ± 0.111	0.831 ± 0.210	<u>1.024 ± 0.264</u>	0.904 ± 0.208	0.741 ± 0.140	0.909 ± 0.208	1.140 ± 0.435
Enhancement Method: KinD								
	GTF	FusionGAN	GANMcC	NestFuse	DenseFuse	SDNet	U2Fusion	Ours
AG	4.567 ± 2.501	2.855 ± 1.246	2.808 ± 1.146	4.659 ± 2.373	3.659 ± 1.713	4.677 ± 2.547	<u>4.874 ± 2.271</u>	4.922 ± 1.479
CC	0.600 ± 0.107	0.625 ± 0.081	0.687 ± 0.082	0.631 ± 0.084	<u>0.698 ± 0.084</u>	0.652 ± 0.097	<u>0.686 ± 0.086</u>	0.716 ± 0.080
EN	7.166 ± 0.278	6.622 ± 0.213	6.857 ± 0.198	7.028 ± 0.274	7.056 ± 0.217	6.967 ± 0.264	<u>7.207 ± 0.188</u>	7.599 ± 0.075
SCD	1.124 ± 0.199	0.757 ± 0.452	1.160 ± 0.347	0.995 ± 0.407	1.306 ± 0.271	1.145 ± 0.279	<u>1.450 ± 0.244</u>	1.687 ± 0.143
SD	9.773 ± 0.743	8.958 ± 0.620	9.743 ± 0.543	9.810 ± 0.574	9.756 ± 0.558	9.407 ± 0.817	10.076 ± 0.643	<u>9.936 ± 0.437</u>
VIF	0.849 ± 0.184	0.617 ± 0.134	0.903 ± 0.243	1.229 ± 0.394	1.034 ± 0.284	0.798 ± 0.173	1.019 ± 0.278	<u>1.140 ± 0.435</u>
Enhancement Methods: EnlightenGAN								
	GTF	FusionGAN	GANMcC	NestFuse	DenseFuse	SDNet	U2Fusion	Ours
AG	4.684 ± 2.183	2.755 ± 1.159	2.808 ± 1.113	4.922 ± 2.214	3.656 ± 1.552	4.864 ± 2.276	5.067 ± 2.074	<u>4.992 ± 1.479</u>
CC	0.601 ± 0.105	0.621 ± 0.081	0.680 ± 0.080	0.597 ± 0.074	<u>0.691 ± 0.081</u>	0.662 ± 0.094	0.684 ± 0.081	0.716 ± 0.080
EN	7.172 ± 0.275	6.623 ± 0.220	6.865 ± 0.164	7.180 ± 0.242	7.130 ± 0.208	7.009 ± 0.257	<u>7.279 ± 0.183</u>	7.599 ± 0.075
SCD	1.130 ± 0.221	0.749 ± 0.460	1.131 ± 0.357	0.949 ± 0.319	1.333 ± 0.226	1.221 ± 0.262	<u>1.452 ± 0.224</u>	1.687 ± 0.143
SD	9.798 ± 0.757	8.873 ± 0.659	9.638 ± 0.549	9.811 ± 0.717	9.846 ± 0.548	9.513 ± 0.781	10.274 ± 0.578	<u>9.936 ± 0.437</u>
VIF	0.882 ± 0.195	0.642 ± 0.140	0.972 ± 0.289	1.460 ± 0.484	<u>1.160 ± 0.337</u>	0.859 ± 0.196	1.153 ± 0.346	<u>1.140 ± 0.435</u>

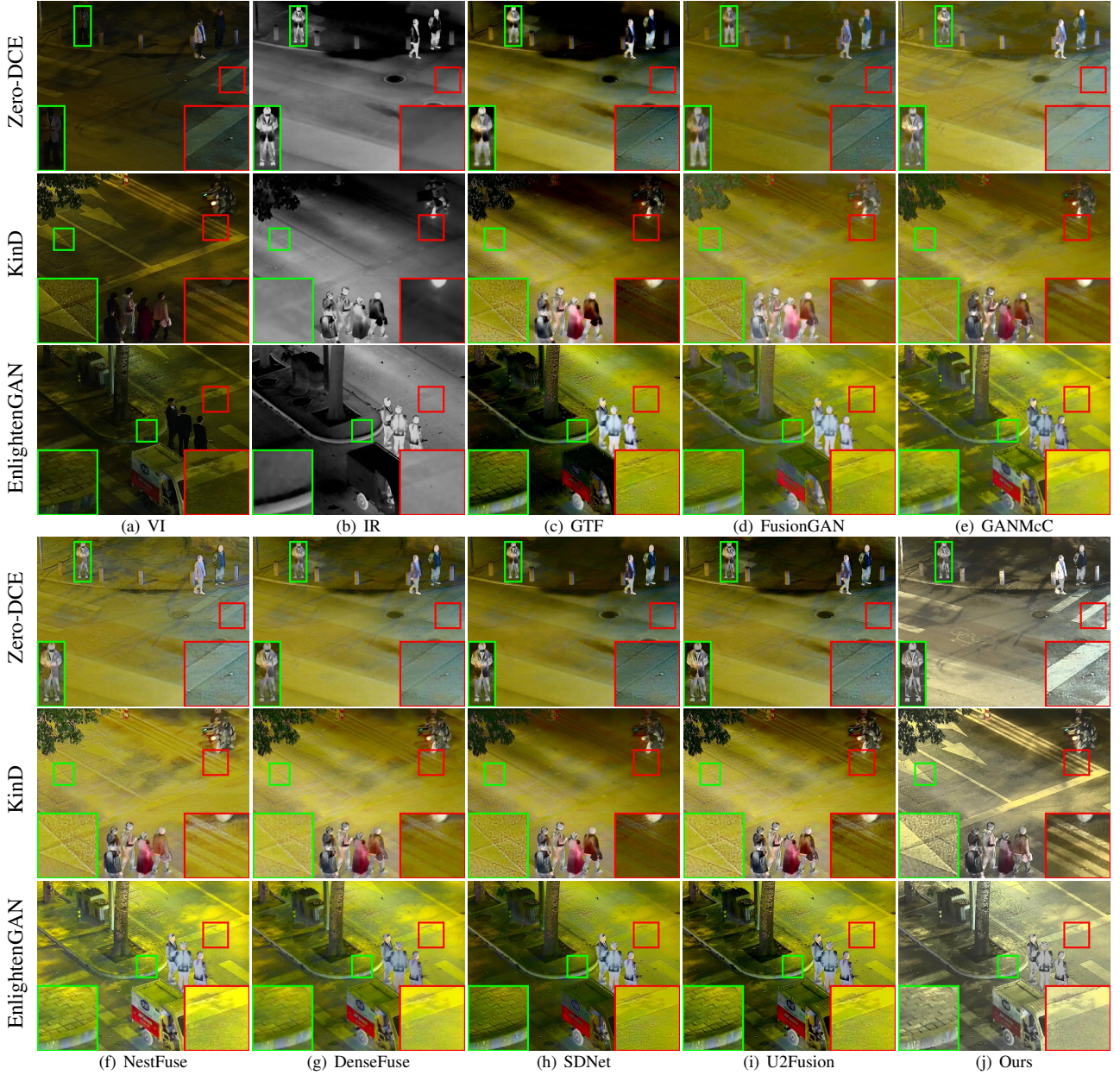


Figure 7: Vision quality comparison of two-stage fusion experiments. Each row represents a scene, and from top to bottom is #21006, #220312, and #260092 images from LLVIP dataset. ((a)-(b): source images, (c)-(i): two-stage fusion results by different enhancement methods and fusion methods, (j): our fusion result).

visible images. Zero-DCE is a curve-based low-light enhancement method that can pixel-wise adjust the dynamic range of the input to obtain an enhanced image. In the first row of Figure 7, we provide two-stage fusion results of one typical image (termed as #21006) from LLVIP dataset to illustrate the distinctive characteristics of our method.

As shown in the red box of scene #21006, GTF, DenseFuse, SDNet, and U2Fusion fail to retain contrast information from the visible image and suffer from spectral contamination in different degrees. Although FusionGAN, GANMcC, and NestFuse integrate the complementary information of source images, the background still lacks texture details.

Only our method boosts the contrast of the whole scene and preserves vivid background. All algorithms retain salient targets from the infrared image in the green box of the #21006 image. However, they all fail to clearly show texture details in targets except for our DIVFusion. Different from other algorithms, our result and the fusion result obtained by Zero-DCE+NestFuse are brighter and have higher contrast. Although the final two-stage fusion result obtained by Zero-DCE+NestFuse is brighter and has more texture detail information, the combination of enhancement and fusion methods neglects the color distortion. On the contrary, our method fully considers color distribution during enhance-

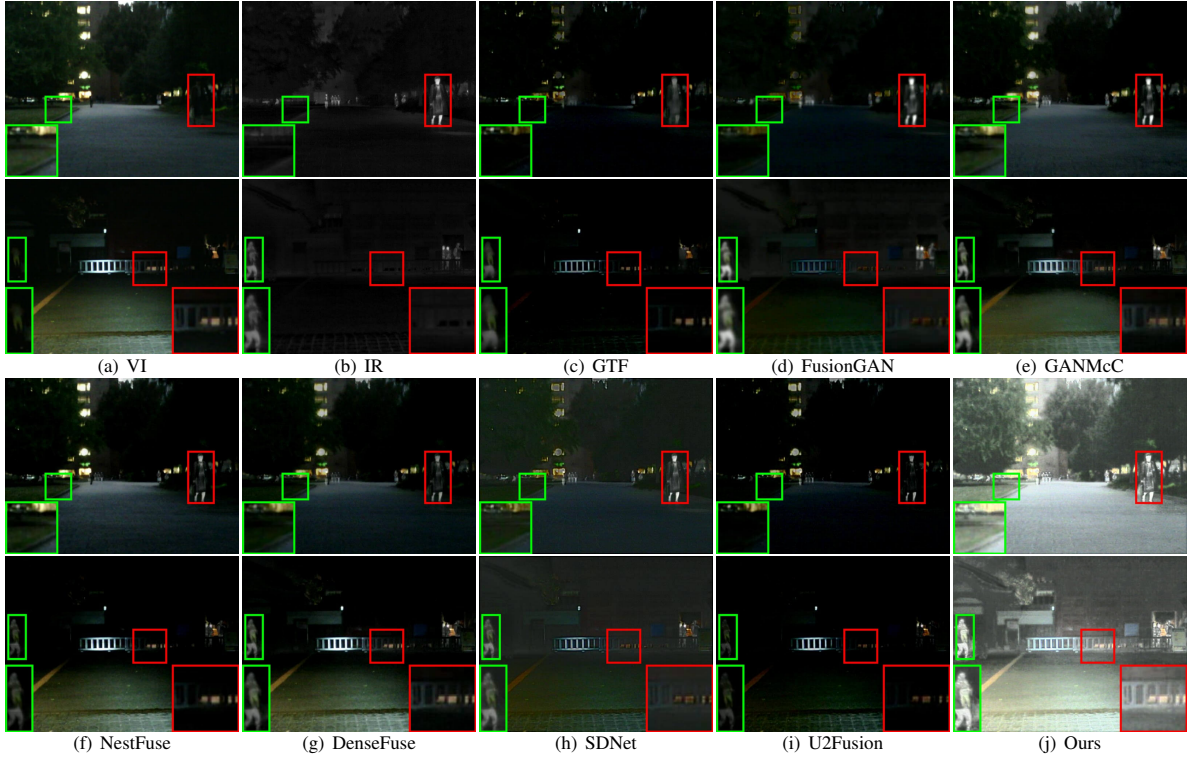


Figure 8: Vision quality comparison of our method with seven SOTA fusion methods on #00004N and #01242N images from MSRS dataset.

ment as well as fusion and provides a natural fused image, benefiting from the color consistency loss.

Quantitative results are presented in Table 1. 50 image pairs are selected from LLVIP dataset to evaluate objectively on AG, CC, EN, SCD, SD, and VIF. Our DIVFusion achieves the largest values in other five metrics and the second place in the SD metric. The best results in AG, EN, and VIF mean that our results have richer texture details, more scene information, and better visual perception, respectively. Our method also performs well in CC and SCD metrics which indicates that our fusion results have higher consistency with source images and are more natural, with the help of the color consistency loss and TCEFNet. For the SD metric, our fused images are slightly lower than those of GTF and U2Fusion, which is justifiable. Specifically, our method sacrifices a little contrast information in order to improve the overall brightness, so as to obtain fused images with a daytime-like scene and better visual perception.

4.4.2. Enhance by KinD

Except for Zero-DCE, we select a Retinex-based low-light enhancement method, *i.e.*, KinD to pre-enhance nighttime visible images. KinD strips the luminance component from the source image to obtain the enhanced result. We choose one typical image (named #220312), which is shown in the second row of Figure 7, to illustrate two-stage fusion results.

As we can see in the second row of Figure 7, combinations of KinD and SOTA fusion methods all provide

brighter scenes, but a lot of contrast information is lost during the enhancement process. For instance, the sidewalk in the original visible image scene fades out during the enhancement and fusion process. By comparison, the crosswalk and sidewalk are clearly in our result, and our fused image is more suitable for the human visual system. Our approach completely integrates complementary of source images and enhancement of nighttime images via the SIDNet and TCEFNet. Therefore, the above-mentioned issues, *i.e.*, illumination degradation, weak contrast information, and color distortion, do not appear in our fusion result.

Quantitative comparisons are also given in Table 1 to better illustrate the advantages of our DIVFusion. We can easily observe that the proposed method performs best in AG, CC, EN, and SCD metrics, slightly lower than NestFuse in the VIF metric, and slightly lower than U2Fusion in the SD metric. As mentioned before, the SD metric mainly measures the contrast information in the whole image. Our method pays more attention to the balance of contrast information and overall brightness, so the SD metric ranks second which is excusable. Moreover, the VIF index of our result is slightly lower than that of NestFuse. However, our DIVFusion does not lose out to NestFuse in the overall performance, because our method reduces redundancy information in the infrared image and injects more effective information from the visible image into the fused image. As a result, our fused image has a better visual perception.

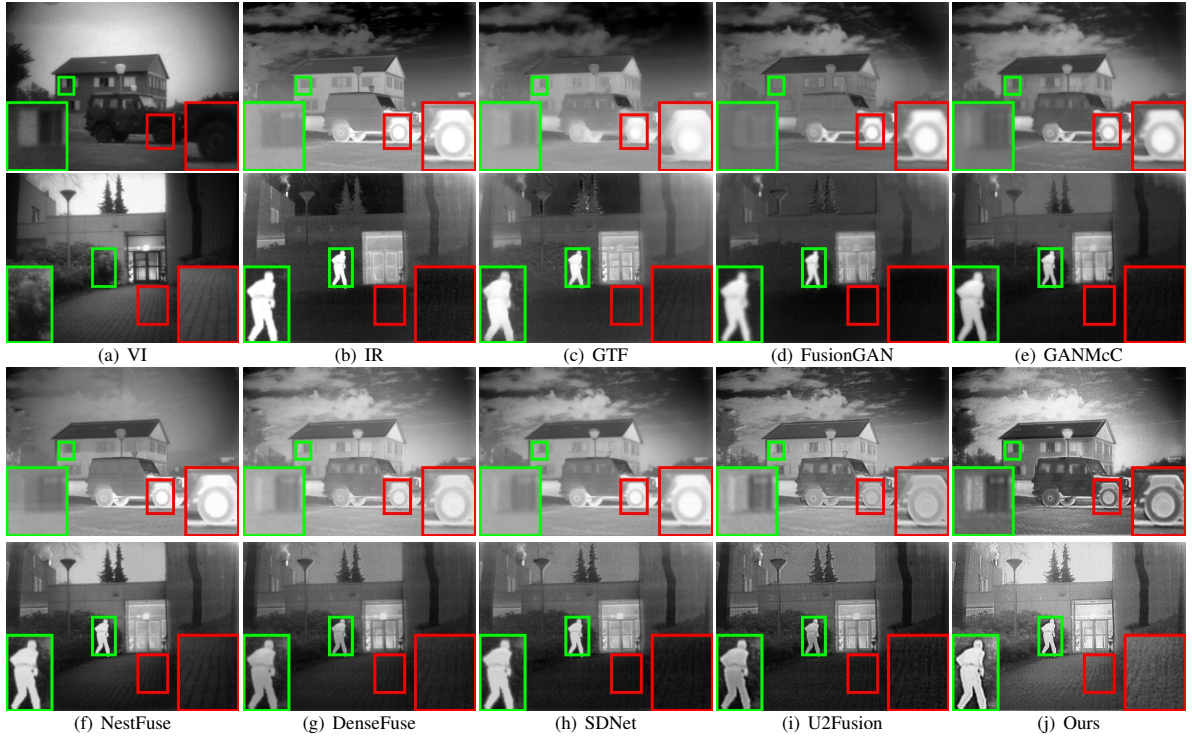


Figure 9: Vision quality comparison of our method with seven SOTA fusion methods on two typical images from TNO dataset.

4.4.3. Enhance by EnlightenGAN

Moreover, we choose an unsupervised GAN-based low-light enhancement method, *i.e.*, EnlightenGAN to implement the pre-enhance process of nighttime visible images. The visualized results are presented in the third row of Figure 7. The typical image shown in Figure 7 is termed as scene #260092 in LLVIP dataset.

As shown in the third row of Figure 7, combinations of EnlightenGAN and SOTA fusion methods lead to serious color distortion. Instead, our method does not have such issues and is more robust to the influence of light sources in images. Specifically, the color consistency loss suppresses color distortion caused by light source amplification in the process of enhancement and fusion. It also shows that the combination of enhancement and fusion methods alone may not achieve the expected results. Our method solves the above-mentioned problem of incompatibility between enhancement and fusion successfully by implementing the enhancement and fusion tasks jointly rather than treating them as two separate components.

From Table 1, one can notice that our DIVFusion performs the best in EN, CC, and SCD metrics. In other words, it demonstrates the superiority of our method in retaining valid information and utilizing interactive information. Our DIVFusion ranks the second in AG metric which lags behind U2Fusion. It means that our method still contains rich texture details. Although the VIF metric of our results is not the largest, our method greatly alleviates the color distortion problem in fused images. On the contrary, fusion results generated by other algorithms, as shown in Figure 7, produce

severe color distortion, which affects the overall quality of fused images.

Overall, it can be observed that the combination of SOTA low-light enhancement methods and fusion methods can achieve nighttime infrared and visible image fusion to some extent. However, this combination leads to severe incompatibility problems between image enhancement and image fusion. Specifically, color distortion and contrast absence are prone to occur in results of simple combination methods. On the contrary, our approach facilitates the nighttime infrared and visible image fusion task through the enhancement process. Specifically, we design an enhancement network, *i.e.*, SIDNet to recover the scene information submerged in the darkness, which boosts the performance of image fusion. Besides, the color consistency loss eliminates the color distortion caused by the above incompatibility problem between enhancement and fusion. Hence, DIVFusion utilizes the enhancement task to facilitate the fusion performance, making both tasks mutually friendly.

4.5. Generalization experiment

The generalization ability of the deep learning-based method also reflects the performance of the model. Therefore, we verify the generalization ability of our method on 50 pairs of nighttime infrared and visible images from MSRS dataset. We also test our method on 30 infrared and visible image pairs from TNO dataset. It is worth noting that our DIVFusion is trained on LLVIP dataset and directly tested on MSRS and TNO datasets.

Table 2

Quantitative results of six metrics, *i.e.*, AG, CC, EN, SCD, SD, and VIF, on 50 image pairs from MSRS dataset.

	GTF	FusionGAN	GANMcC	DenseFuse	NestFuse	SDNet	U2Fusion	Ours
AG	1.405 ± 0.297	1.04 ± 0.207	1.613 ± 0.343	1.729 ± 0.381	<u>2.191 ± 0.531</u>	1.817 ± 0.385	1.434 ± 0.436	5.572 ± 0.717
CC	0.613 ± 0.074	0.655 ± 0.046	<u>0.709 ± 0.055</u>	0.727 ± 0.06	0.704 ± 0.067	0.663 ± 0.056	0.691 ± 0.056	0.609 ± 0.05
EN	5.171 ± 0.389	5.13 ± 0.334	<u>5.67 ± 0.327</u>	5.354 ± 0.391	<u>5.691 ± 0.394</u>	4.671 ± 0.381	4.101 ± 0.565	7.319 ± 0.092
SCD	0.951 ± 0.248	0.922 ± 0.312	<u>1.445 ± 0.211</u>	1.373 ± 0.173	1.665 ± 0.134	0.882 ± 0.17	1.047 ± 0.151	1.348 ± 0.117
SD	5.715 ± 0.755	5.22 ± 0.711	<u>6.986 ± 0.795</u>	5.811 ± 0.838	6.343 ± 0.778	4.622 ± 0.632	4.265 ± 0.806	9.936 ± 0.412
VIF	0.542 ± 0.065	0.492 ± 0.135	<u>0.676 ± 0.085</u>	0.715 ± 0.069	<u>0.883 ± 0.086</u>	0.414 ± 0.122	0.434 ± 0.095	0.946 ± 0.135

Table 3

Quantitative results of six metrics, *i.e.*, AG, CC, EN, SCD, SD, and VIF, on 30 image pairs from TNO dataset.

	GTF	FusionGAN	GANMcC	DenseFuse	NestFuse	SDNet	U2Fusion	Ours
AG	2.954 ± 1.305	2.399 ± 0.768	2.537 ± 0.734	3.394 ± 0.877	3.596 ± 0.997	3.420 ± 1.048	<u>4.749 ± 1.223</u>	6.266 ± 1.396
CC	0.383 ± 0.259	0.461 ± 0.198	<u>0.544 ± 0.202</u>	0.558 ± 0.211	0.504 ± 0.227	0.518 ± 0.193	0.543 ± 0.213	0.527 ± 0.220
EN	6.947 ± 0.431	6.677 ± 0.434	6.849 ± 0.394	6.884 ± 0.359	<u>7.087 ± 0.299</u>	7.058 ± 0.339	7.071 ± 0.330	7.543 ± 0.195
SCD	0.911 ± 0.272	1.163 ± 0.845	1.541 ± 0.612	1.748 ± 0.385	1.611 ± 0.323	1.502 ± 0.745	1.742 ± 0.391	<u>1.745 ± 0.208</u>
SD	10.171 ± 1.499	9.327 ± 1.309	9.345 ± 0.890	9.627 ± 1.106	9.729 ± 0.930	9.919 ± 0.936	<u>9.987 ± 0.991</u>	9.802 ± 0.547
VIF	0.696 ± 0.180	0.663 ± 0.127	0.767 ± 0.149	0.897 ± 0.195	1.042 ± 0.189	0.912 ± 0.179	0.888 ± 0.185	<u>0.923 ± 0.183</u>

4.5.1. Qualitative results of MSRS dataset

Qualitative comparison results on MSRS dataset are shown in Figure 8. Our method is able to provide a brighter scene with salient targets, and the information submerged in darkness is tapped. As shown in the red box of scene #00004N, our method improves the overall image contrast, so the salient target is remarkably improved compared with the original infrared image. It is worth emphasizing that our color consistency loss effectively maintains the color fidelity of the fused image compared with other alternatives. The difference can also be easily seen in the green box of the #01242N image. Other fusion algorithms have no way to provide this region of scene information because the original visible image is too dark, making it difficult to represent details in the fused image. In the #01242N image, our method taps into a lot of scene information hidden in the darkness. For example, the building in the upper left corner contains both high contrasts from infrared images and rich texture details from visible images.

4.5.2. Quantitative results of MSRS dataset

Comparative results of different methods on six metrics are presented in Table 2. Our method ranks first in AG, EN, SD, and VIF metrics by a significant margin on MSRS dataset. The best results in AG and EN mean that our method captures abundant texture details and contains rich information. The optimal results on SD and VIF indicate that our results obtain the most obvious contrast and the best visual perceptual performance, respectively. For the CC metric, the correlation coefficient is lower because of the stripping illumination operation, which reduces the linear correlation between the fused image and source images. Regarding the SCD metric, our method enhances the overall brightness, reducing the similarity between the nighttime visible image as well as the differential image. It is worth

noting that our results can obtain brighter scenes, and the enhancement and fusion operations promote each other to produce a high-quality fused image with better visual perception. In summary, our model, trained on LLVIP dataset, also shows vivid fusion results on MSRS dataset.

4.5.3. Qualitative results of TNO dataset

We also perform quantitative and qualitative comparisons on TNO dataset to demonstrate the generalization performance of our method. It is worth noting that the infrared and visible image pairs in TNO dataset are grayscale images and thus our framework cannot take full advantage of its potential. However, our method still exhibits comparable performances to SOTA fusion methods. As shown in Figure 9, fusion results of GTF, FusionGAN, and GANMcC are relatively blurred, resulting in a large amount of missing scene information. Although NestFuse, DenseFuse, SDNet, and U2Fusion have more texture information, the overall scene submerges into the darkness and fails to provide a pleasing visual perception. On the contrary, our method can not only provide abundant texture information and thermal targets, but it also lights up the whole scene with high contrast.

4.5.4. Quantitative results of TNO dataset

The comparative results of different approaches on TNO dataset are presented in Table 3. From the results, we can see that our framework ranks first in AG and EN metrics, which indicates that our model can sufficiently maintain texture information in source images and contain rich information. Although our method ranges the sub-optimal performance on the SCD metric, the margin from the best one is tiny. It demonstrates that our fusion results introduce fewer artifacts. Besides, DIVFusion has an excellent performance on the VIF metric only following NestFuse, revealing that

Table 4

Quantitative evaluation results of ablation study on 50 pairs of infrared and visible images from LLVIP dataset.

	W/O $\mathcal{L}_{tex}$	W/O $\mathcal{L}_{color}$	W/O GRM	W/O CEM	W/O SIDNet	Ours
AG	4.359 ± 1.608	3.153 ± 0.734	4.287 ± 1.310	4.401 ± 1.319	4.102 ± 1.294	4.922 ± 1.479
CC	0.707 ± 0.096	0.493 ± 0.087	0.703 ± 0.095	0.676 ± 0.104	0.664 ± 0.100	0.716 ± 0.080
EN	7.493 ± 0.106	7.433 ± 0.099	7.436 ± 0.096	7.336 ± 0.156	7.574 ± 0.075	7.599 ± 0.075
SCD	1.790 ± 0.157	0.549 ± 0.280	1.659 ± 0.191	1.569 ± 0.214	1.631 ± 0.161	1.687 ± 0.143
SD	10.081 ± 0.585	10.045 ± 0.489	10.218 ± 0.571	9.840 ± 0.607	10.337 ± 0.483	9.936 ± 0.437
VIF	1.062 ± 0.218	0.622 ± 0.139	1.028 ± 0.217	0.905 ± 0.153	1.022 ± 0.253	1.140 ± 0.435

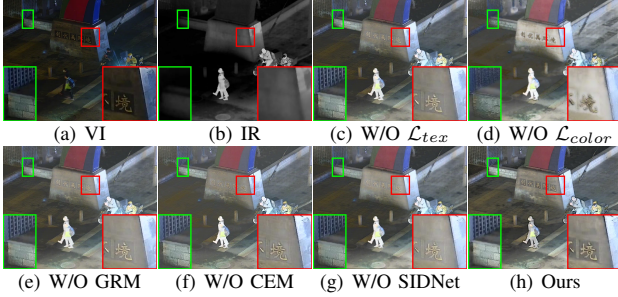


Figure 10: Vision quality comparison of the ablation study on important loss functions and modules on #010175 image from LLVIP dataset.

our fused images are also consistent with the human visual system. For the CC metric, our fusion results obtain brighter scenes, which leads to inconsistency of linear correlation. Hence, it is excusable that the CC metric ranks fourth. Our approach attempts to balance the enhancement and high contrast, leading to contrast reduction in some places. Thus, the SD metric ranks fourth. Notably, our color consistency loss cannot be exploited in TNO dataset and thus our DIVFusion fails to take full advantage of its potential. Nonetheless, the performance of our method still achieves comparable among SOTA methods.

4.6. Ablation study

4.6.1. Texture loss analysis

The texture loss guides the network to generate fused images with the maximum textures of infrared and visible images. After the texture loss is removed, fused images fail to retain enough texture details. As shown in Figure 10(c), the text on the stone sign in the scene is indistinguishable. In addition, the structural information of infrared images is also not reflected in fused images at nighttime.

4.6.2. Color consistency loss analysis

The color consistency loss is one of the most important losses in our DIVFusion. On the one hand, the color consistency loss guides the network to inject more scene and texture information of the visible image into the fused image. On the other hand, it guides the fused image to gain a color-balanced scene. As shown in Figure 10(d), the fusion result generated by the proposed method without the color consistency loss is more inclined toward the infrared image,

and the whole image also has obvious color degradation. By contrast, our fusion method simultaneously maintains the balanced color distribution of the whole scene and preserves the texture details of background areas.

4.6.3. Gradient retention module analysis

Our DIVFusion could boost the texture information in the fused images with the guidance of GRM. When we remove the gradient retention module, as shown in Figure 10(e), the overall scene is relatively smoother with less gradient variation and less vivid texture structure. It also indicates that the texture details of the infrared and visible images are not well incorporated into the fused image during the fusion process. It is worth noting that dual residual streams are employed in GRM to progressively complete slight gradient enhancement at the feature level. Hence, removing GRM makes the fusion result miss more micro texture details, which is different from the lack of texture loss.

4.6.4. Contrast enhancement module analysis

Another critical component in our network is CEM, which reinforces the contrast information in mixed features. As shown in Figure 10(f), we can notice that the zebra line and sidewalk are not prominent which provides extremely weak contrast information. Besides, the text on the stone in the scene can be seen in the outline but can not be well identified. On the contrary, our fusion result has a high-contrast scene with good visual perception as shown in Figure 10(h).

4.6.5. Scene-illumination disentangled network analysis

We propose a SIDNet to remove the degenerative illumination from mixed features in order to achieve image enhancement and boost the following fusion process. Without SIDNet, it is difficult for the fused image to express complete scene information, because abundant details are drowned in darkness. In Figure 10(g), the left part of the image hardly supplies effective information, *i.e.*, the crosswalk is hidden in the darkness and hard to identify. On the contrary, in Figure 10(h), the fused image generated by the proposed method with SIDNet is brighter and presents more detailed textures in dark areas. Therefore, we can conclude that SIDNet provides a brighter scene for the subsequent fusion.

Table 5
Computational efficiency comparison with seven SOTA fusion methods.

Method	GTF	FusionGAN	GANMcC	NestFuse	DenseFuse	SDNet	U2Fusion	Ours
Parameters(M)	-	0.926	1.864	2.733	<u>0.074</u>	0.067	0.659	4.403
FLOPs(G)	-	2440.901	4915.995	-	<u>231.045</u>	176.259	1728.740	14454.911
FPS	0.025	1.917	0.942	2.112	<u>2.827</u>	9.657	2.645	0.331

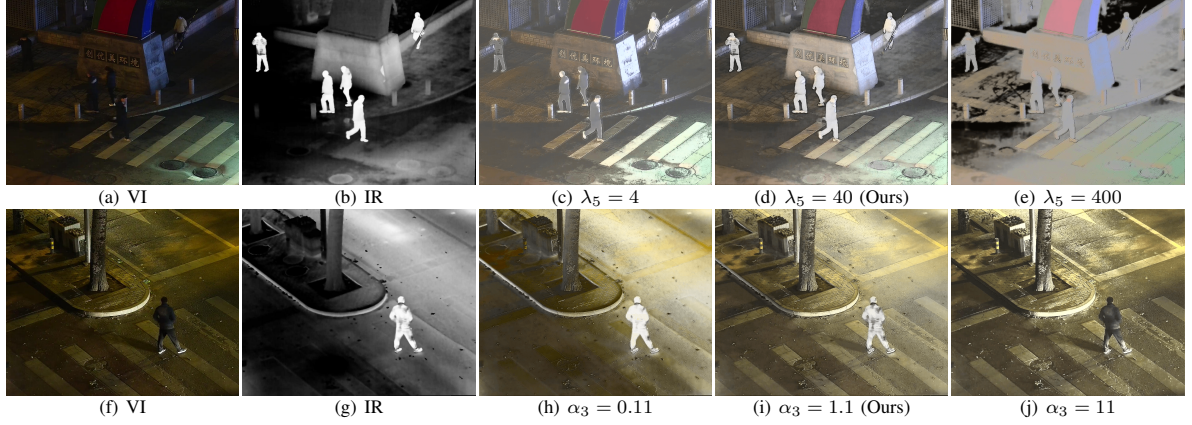


Figure 11: The analysis of hyperparameters. First row: infrared image, visible image, and fusion results with different λ_5 . Second row: infrared image, visible image, and fusion results with different α_3

The qualitative results of ablation studies are shown in Table 4. We can notice that our DIVFusion presents remarkable superiority in AG, CC, EN, and VIF. The SD metric of our method is not the best, because it indicates that our method focuses on the balance of contrast information and overall brightness. It can be observed that our method ranks second in the SCD metric, which lags behind the method without texture loss. However, our DIVFusion captures more slight texture details rather than introducing pseudo-information.

4.7. Computational complexity analysis

As shown in Table 5, we provide training parameters, floating-point operations per second (FLOPs), and frames per second (FPS) of seven SOTA image fusion methods and our DIVFusion. In Table 5, deep learning-based methods achieve higher speed than traditional methods, benefiting from the strong computing ability of GPU. From the results, we can observe that SDNet achieves the minimum parameters, FLOPs, and maximum FPS. This is because SDNet only utilizes the light network architecture in the testing phase. Our DIVFusion contains several attention blocks with residual connections to generate high-contrast and daytime-like fusion images with rich texture. Hence, it is excusable that our method lags slightly behind other deep learning-based methods in terms of computational complexity.

4.8. Analysis of some hyperparameters

4.8.1. Sensitivity analysis of λ_5

As described in Section 3.3.1, λ_5 is the parameter controlling the degree of enhancement of nighttime visible image features. According to Eq. (13), the larger λ_5 value pays more attention to the enhancement process. Since the quality of the enhanced visible features directly affects the generation of fusion results, we assign a large value to λ_5 . However, excessive λ_5 leads to over-exposure in enhanced visible images. In order to analyze the sensitivity of this parameter, λ_5 is set as 4, 40, and 400, respectively. Some vision quality results are shown in the first row of Figure 11.

It can be seen that as λ_5 sets as 4 or 400, the enhancement of fusion images changes from dark to over-exposure. When $\lambda_5 = 40$, our fusion result achieves great visual perception. At this time, our DIVFusion can produce high-quality images with significant contrast and daytime-like scenes.

4.8.2. Sensitivity analysis of α_3

α_3 is used to balance the color correction level and control the injection of the visible information. In order to illustrate the effect of different α_3 on our fusion results. The vision results are displayed in the second row of Figure 11.

It can be found that as α_3 increases, the texture information in fused images is retained better, but the contrast of thermal targets is gradually decreasing. At the same time, the color distortion caused by light source amplification in the process of enhancement and fusion is reduced rapidly. When $\alpha_3 = 1.1$, the preservation of thermal targets and texture details achieves a good balance. The color distribution of fused images can also be adjusted in a great way.

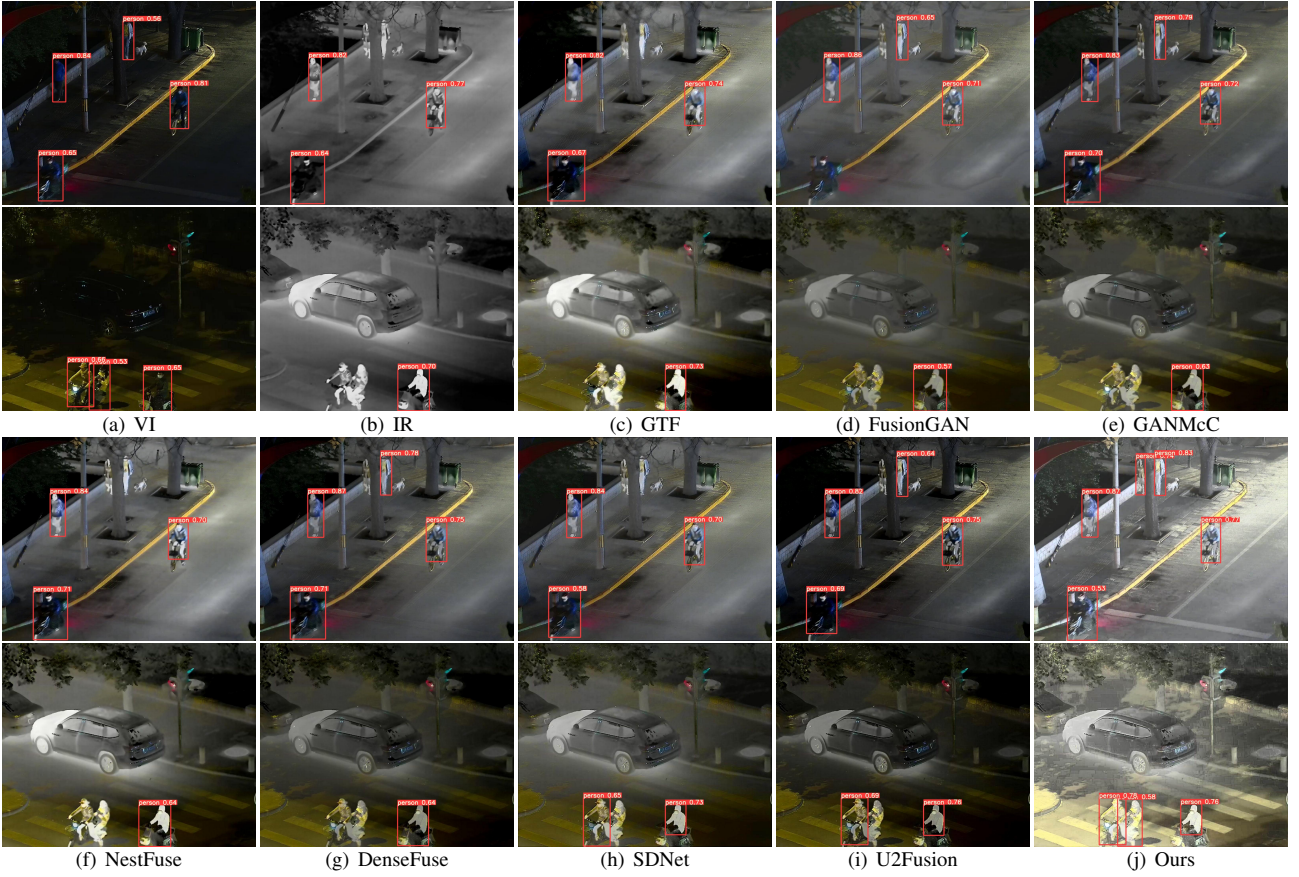


Figure 12: Detection performance of our results with seven SOTA fusion results on #040086 and #250409 images from LLVIP dataset.

Table 6

Pedestrian detection performance for visible, infrared, and fused images on LLVIP dataset.

	Precision	Recall	mAP@0.50	mAP@0.60	mAP@0.70	mAP@0.80	mAP@0.90	mAP@[0.5:0.95]
VI	0.928	0.709	0.810	0.623	0.478	0.230	0.034	0.399
IR	0.921	0.703	0.772	0.663	0.521	0.331	0.077	0.436
GTF	0.887	0.750	0.808	0.714	0.536	0.364	0.072	0.445
FusionGAN	0.858	0.776	0.838	0.752	0.589	0.364	0.081	0.481
GANMcC	0.921	0.797	0.878	0.760	0.578	0.413	0.096	0.498
NestFuse	0.894	0.810	0.869	0.759	0.599	0.382	0.089	0.497
DenseFuse	0.895	0.809	0.882	0.766	0.608	0.393	0.102	0.504
SDNet	0.914	0.797	0.866	0.778	0.586	0.404	0.097	0.508
U2Fusion	0.859	0.823	0.871	0.714	0.551	0.398	0.097	0.476
Ours	0.935	0.857	0.898	0.808	0.627	0.387	0.091	0.520

Therefore, to obtain good fusion performance in experiments, λ_5 and α_3 are set as 40 and 1.1, respectively.

4.9. Detection performance

Our DIVFusion fully exploits the information hidden in the darkness and generates fused images with high-contrast, which can promote the improvement of the performance of high-level vision tasks (*e.g.*, object detection). To further demonstrate that our method can effectively facilitate high-level computer vision tasks, we apply fusion results to a typical high-level computer vision task, *i.e.*, pedestrian

detection. The SOTA detection model YOLOv5 [38] is deployed to conduct pedestrian detection on infrared images, nighttime visible images, and fused images.

4.9.1. Qualitative results

We supply some typical examples in Figure 12 to illustrate the advantages of our joint enhancement and fusion method in facilitating pedestrian detection. Infrared images, visible images, and fusion results generated by fusion methods are directly fed into the YOLOv5 detector, respectively. In the #040086 image, the visible image cannot accentuate

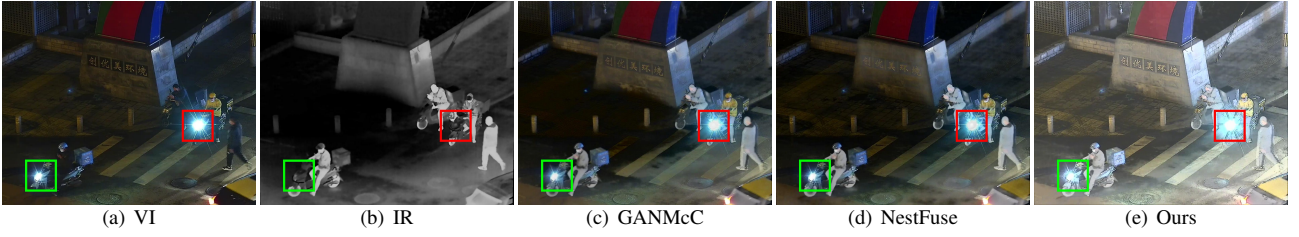


Figure 13: A failure case. Exposure degradation appears in fused images.

all pedestrians effectively because of the illumination degradation, so some pedestrians are lost in the detection result. The infrared image fails to represent enough texture details on pedestrians, so the detection result is disappointing. As for fused images, the complementary information including high contrast, abundant texture, and salient targets helps to detect pedestrians effectively. However, the detector fails to detect pedestrians from fusion results generated by other methods due to the interference of illumination degradation. Low-light environment weakens the scene information of fused images and thus affects detection performance. Hence, pedestrians are still failed to distinguish. Fortunately, DIVFusion overcomes the darkness and fully integrates the complementary information of infrared and visible images, so pedestrians are all detected successfully. In the #250409 scene, fusion results of GTF, NestFuse, and SDNet obtain bright scenes from the infrared image which makes fused images inclined to enhanced infrared image. Nevertheless, the detection performance of pedestrians is still disappointing. The rest fusion methods suffer from severe darkness, which weakens the complementary information of source images. It is worth emphasizing that our method can defend against interference from darkness, and achieve texture preservation and contrast improvement. These characteristics undoubtedly make fused images obtain daytime-like scenes and rich semantic information which is friendly to pedestrian detection.

4.9.2. Quantitative results

We further use quantitative metrics to measure the pedestrian detection task, which is shown in Table 6. Precision denotes the probability that all samples with positive predictions are true samples. A bigger precision means more positive samples are detected. Recall measures the probability of a positive sample being predicted from an actual positive sample. A larger recall indicates that fewer detection results are missed. Another metric for reflecting detection performance is mean average precision (mAP). The more the value of mAP increases close to 1, the better the pedestrian detection is. The mAP@0.50, mAP@0.6, mAP@0.7, mAP@0.8, and mAP@0.9 imply the mAP values at confidence thresholds of 0.5, 0.6, 0.7, 0.8, and 0.9, respectively. The mAP@[0.5:0.95] denotes the average of all mAP values at different IoU thresholds (from 0.5 to 0.95 in steps of 0.05). In Table 6, our fusion results have the highest values in both Precision and Recall, which

indicates that pedestrians detected are most positive and cause fewer misjudgments, respectively. Besides, we also achieve obvious advantages in mAP, which shows higher detection accuracy in our fusion results.

In brief, due to the powerful combination of SIDNet and TCEFNet, our DIVFusion fully combines the salient thermal object information in infrared images with the texture structure information in visible images and improves the overall brightness at the same time. These characteristics in fused images stably boost discrimination of targets, which achieves a better performance in the pedestrian detection task.

5. Conclusion and discussion

In this paper, we propose a novel nighttime infrared and visible image fusion algorithm based on visual enhancement, which achieves coupling and mutual benefiting of image fusion and image enhancement. Specifically, SIDNet is designed to separate degenerate illumination features from mixed features and avoid vital information of source images loss. Then, we employ TCEFNet which contains two special modules to obtain contrast enhancement and texture retention during the fusion process. Considering the color distortion due to the incompatibility of enhancement and fusion tasks, we design a color consistency loss to adjust the color distribution of fused images. Qualitatively and quantitatively comparisons with state-of-the-art methods validate the superiority of our method, including visual perception, scene brightness, and complementary information integration. Two-stage fusion experiments are also performed to illustrate that our method effectively helps to mitigate the incompatibility between fusion and enhancement tasks. Moreover, pedestrian detection experiments demonstrate the potential of our DIVFusion for high-level computer vision tasks.

The limitation of our method, also of the other methods, is the poor ability to solve exposure degradation. We provide a typical example to illustrate this intuitively, as shown in Figure 13. Although all fusion methods fail to eliminate over-exposed effects in the headlight region, our method provides more natural visual perception in other regions than other algorithms. A possible solution is to model point light sources by Gaussian distribution and design a decomposition network to strip the over-exposure map from the visible image. Due to the absence of effective information in the visible image in over-exposure regions,

we will integrate more information to fix over-exposure areas from the infrared image in the fusion process. In the future, we will further devise an illuminance adjustment module to address both the low-light and over-exposed degradation in infrared and visible image fusion tasks.

CRedit authorship contribution statement

Lin Feng Tang: Methodology, Experiment, Writing. **Xinyu Xiang:** Methodology, Experiment, Writing. **Hao Zhang:** Methodology. **Meiqi Gong:** Experiment. **Jiayi Ma:** Conceptualization of this study, Methodology, Revised the manuscript

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